

Name - S. Jaxon Rizal
Reg. No - 192511421
Course Code - CSA0313

ASSESSMENT TOOL 2 - SHORT ANSWER TEST

COURSE NAME : Data Structures

COURSE CODE : CSA0313

QUESTIONS: EACH QUESTION CARRIES: 1 MARKS

1. What is a graph without cycles called?

Ans: Acyclic graph.

2. What do you call a graph where every vertex is reachable from every other vertex?

Ans: Connected graph.

3. Which graph representation uses a two-dimensional array?

Ans: Adjacency matrix.

4. Which graph structure is ideal for task scheduling problems?

Ans: Directed Acyclic Graph (DAG).

5. What type of graph models one-way road systems?

Ans: Directed graph.

6. What structure is obtained when all cycles are removed from a graph?

Ans: Tree.

7. Which graph model is used to represent dependencies?

Ans: Directed Acyclic Graph (DAG).

8. What key property must a graph satisfy before applying topological sorting?

Ans: The graph must be a Directed Acyclic Graph (DAG).

9. Which graph type is commonly used in job assignment problems?

Ans: Bipartite graph.

10. What graph structure represents hierarchical relationships?

Ans: Tree.

11. For which type of graph does topological ordering exist?

Ans: Directed Acyclic Graph (DAG).

12. If some vertices are not processed in topological sorting, what does it indicate?

Ans: It indicates the presence of a cycle in the graph.

13. Which data structure is used to store results in DFS-based topological sorting?

Ans: Stack.

14. Which technique is commonly used to detect cycles in topological sorting?

Ans: Depth First Search (DFS).

15. Which graph model is suitable for course prerequisite systems?

Ans: Directed Acyclic Graph (DAG).

16. Which data structure improves node selection in Kahn's algorithm?

Ans: Queue.

17. What type of graph is used in compiler build systems?

Ans: Directed Acyclic Graph (DAG).

18. Which MST algorithm expands by selecting edges from a starting vertex?

Ans: Prim's algorithm.

19. Which shortest path algorithm can detect negative cycles?

Ans: Bellman-Ford algorithm.

20. Which traversal is used to find shortest paths in unweighted graphs?

Ans: Breadth First Search (BFS).

21. What key operation is repeatedly used in shortest path algorithms?
Ans: Relaxation.
22. Which algorithm applies a greedy strategy for shortest paths?
Ans: Dijkstra's algorithm.
23. Which algorithm uses dynamic programming for shortest paths?
Ans: Floyd-Warshall algorithm.
24. What is the time complexity of the Floyd-Warshall algorithm?
Ans: $O(V^3)$.
25. What data structure is used in Floyd-Warshall to store distances?
Ans: Two-dimensional distance matrix.
26. Which traversal technique relies on recursion or stack?
Ans: Depth First Search (DFS).
27. What is the process of visiting all vertices in a graph called?
Ans: Graph traversal.
28. Which concept is applied in network design optimization problems?
Ans: Minimum Spanning Tree (MST).
29. Removing the heaviest edge from a cycle ensures what property?
Ans: It prevents the formation of cycles.
30. Which property guarantees the correctness of an MST?
Ans: Cut Property.

31. Which algorithm computes all-pairs shortest paths?

Ans: Floyd-Warshall algorithm.

32. What is the core concept behind repeated updates in shortest path algorithms?

Ans: Relaxation.

33. Which traversal is used to check if a path exists between two nodes?

Ans: Depth First Search (DFS) / Breadth First Search (BFS).

34. Which strategy is used in uninformed AI search?

Ans: Blind Search.

35. Which traversal guarantees shortest path in unweighted graphs?

Ans: Breadth First Search (BFS).

36. What is another term for a vertex?

Ans: Node.

37. Which traversal approach is used in Kahn's algorithm?

Ans: Topological Sort (BFS-based).

38. Which method for topological sorting uses recursion?

Ans: DFS-based Topological Sort.

39. What is a sequence of vertices from source to destination called?

Ans: Path.

40. What type of graph has weighted edges?

Ans: Weighted graph.